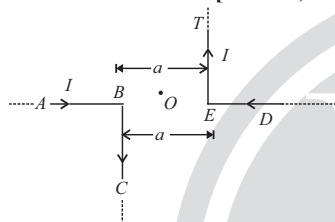


Moving Charges and Magnetism

Single Correct Type Questions

1. The magnitude of magnetic induction at mid-point O due to current arrangement as shown in the Fig will be:

[29 Jan, 2023 (Shift-I)]



- (a) $\frac{\mu_0 I}{2\pi a}$ (b) 0
(c) $\frac{\mu_0 I}{4\pi a}$ (d) $\frac{\mu_0 I}{\pi a}$

2. Match List-I with List-II

[25 Jan, 2023 (Shift-I)]

List - I (Current configuration)	List - II (Magnetic field at point O)
A	I. $B_0 = \frac{\mu_0 I}{4\pi r} [\pi + 2]$
B	II. $B_0 = \frac{\mu_0 I}{4 r}$
C	III. $B_0 = \frac{\mu_0 I}{2\pi r} [\pi - 1]$
D	IV. $B_0 = \frac{\mu_0 I}{4\pi r} [\pi + 1]$

Choose the correct answer from the option given below:

- (a) A-III, B-IV, C-I, D-II (b) A-I, B-III, C-IV, D-II
(c) A-III, B-I, C-IV, D-II (d) A-II, B-I, C-IV, D-III

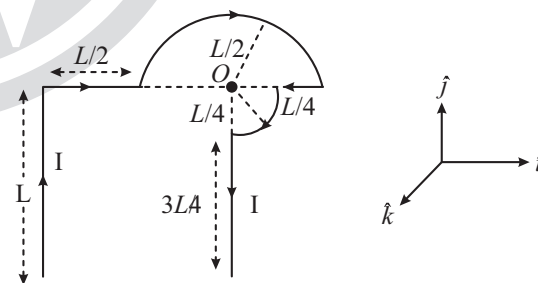
3. The magnetic field at the centre of a circular coil of radius r , due to current I flowing through it, is B . The magnetic field at a point along the axis at a distance $r/2$ from the centre is:

[24 June, 2022 (Shift-I)]

- (a) $B/2$ (b) $2B$
(c) $\left(\frac{2}{\sqrt{5}}\right)^3 B$ (d) $\left(\frac{1}{\sqrt{3}}\right)^3 B$

4. Which one of the following options represents the magnetic field $\vec{B}$ at O due to the current flowing in the given wire segments lying on the xy plane?

[JEE Adv, 2022]

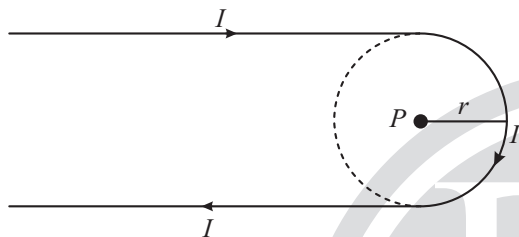


- (a) $\vec{B} = \frac{-\mu_0 I}{L} \left(\frac{3}{2} + \frac{1}{4\sqrt{2}\pi} \right) \hat{k}$
(b) $\vec{B} = -\frac{\mu_0 I}{L} \left(\frac{3}{2} + \frac{1}{2\sqrt{2}\pi} \right) \hat{k}$
(c) $\vec{B} = \frac{-\mu_0 I}{L} \left(1 + \frac{1}{4\sqrt{2}\pi} \right) \hat{k}$
(d) $\vec{B} = \frac{-\mu_0 I}{L} \left(1 + \frac{1}{4\pi} \right) \hat{k}$

5. A coil having N turns is wound tightly in the form of a spiral with inner and outer radii ' a ' and ' b ' respectively. Find the magnetic field at centre, when a current I passes through coil: [31 Aug, 2021 (Shift-I)]

(a) $\frac{\mu_0 I}{8} \left[\frac{a+b}{a-b} \right]$ (b) $\frac{\mu_0 IN}{2(b-a)} \log_e \left(\frac{b}{a} \right)$
 (c) $\frac{\mu_0 I}{4(a-b)} \left[\frac{1}{a} - \frac{1}{b} \right]$ (d) $\frac{\mu_0 I}{8} \left(\frac{a-b}{a+b} \right)$

6. A hairpin like shape as shown in figure is made by bending a long current carrying wire. What is the magnitude of a magnetic field at point P which lies on the centre of the semicircle? [17 March, 2021 (Shift-II)]

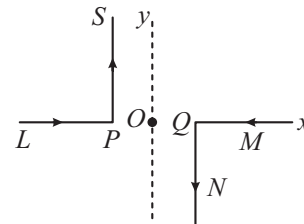


(a) $\frac{\mu_0 I}{2\pi r} (2 - \pi)$ (b) $\frac{\mu_0 I}{4\pi r} (2 - \pi)$
 (c) $\frac{\mu_0 I}{2\pi r} (2 + \pi)$ (d) $\frac{\mu_0 I}{4\pi r} (2 + \pi)$

7. A proton and an α -particle (with their masses in the ratio of 1 : 4 and charges in the ratio of 1 : 2) are accelerated from rest through a potential difference V . If a uniform magnetic field (B) is set up perpendicular to their velocities, the ratio of the radii $r_p : r_\alpha$ of the circular paths described by them will be [12 Jan, 2019 (Shift-I)]

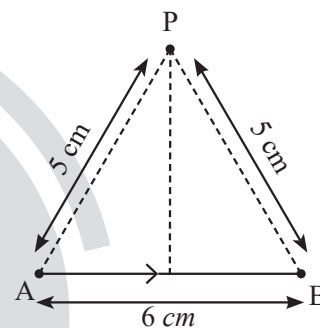
(a) $1 : \sqrt{2}$ (b) $1 : 2$
 (c) $1 : 3$ (d) $1 : \sqrt{3}$

8. As shown in the figure, two infinitely long, identical wires are bent by 90° and placed in such a way that the segments LP and QM are along the x -axis, while segments PS and QN are parallel to the y -axis. If $OP = OQ = 4$ cm, and the magnitude of the magnetic field at O is 10^{-4} T, and the two wires carry equal current (see figure), the magnitude of the current in each wire and the direction of the magnetic field at O will be ($\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$) [12 Jan, 2019 (Shift-I)]



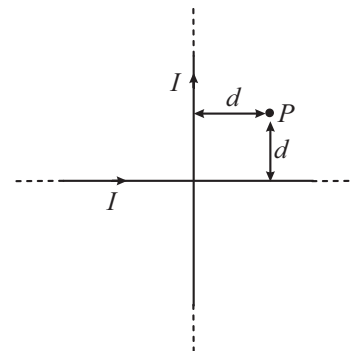
- (a) 20 A, perpendicular out of the page.
 (b) 40 A, perpendicular out of the page.
 (c) 20 A, perpendicular into the page.
 (d) 40 A, perpendicular into the page.

9. Find the magnetic field at point P due to a straight line segment AB of length 6 cm carrying a current of 5 A. (See figure) ($\mu_0 = 4\pi \times 10^{-7} \text{ N-A}^{-2}$) [12 April, 2019 (Shift-II)]



- (a) $2.0 \times 10^{-5} \text{ T}$ (b) $3.0 \times 10^{-5} \text{ T}$
 (c) $2.5 \times 10^{-5} \text{ T}$ (d) $1.5 \times 10^{-5} \text{ T}$

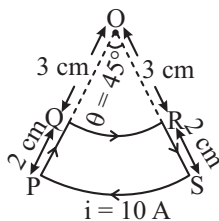
10. Two very long, straight and insulated wires are kept at 90° angle from each other in xy -plane as shown in the figure. These wires carry currents of equal magnitude I , whose directions are shown in the figure. The net magnetic field at point P will be [8 April, 2019 (Shift-II)]



- (a) Zero (b) $\frac{+\mu_0 I}{\pi d} (\hat{z})$
 (c) $-\frac{\mu_0 I}{2\pi d} (\hat{x} + \hat{y})$ (d) $\frac{\mu_0 I}{2\pi d} (\hat{x} + \hat{y})$

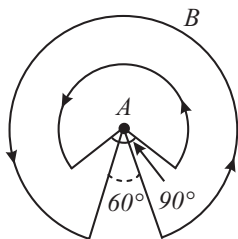
11. A current loop, having two circular arcs joined by two radial lines is shown in the figure. It carries a current of 10 A. The magnetic field at point O will be close to:

[9 Jan, 2019 (Shift-I)]



- (a) $1.0 \times 10^{-7} \text{ T}$ (b) $1.5 \times 10^{-7} \text{ T}$
 (c) $1.5 \times 10^{-5} \text{ T}$ (d) $1.0 \times 10^{-5} \text{ T}$
12. A long solenoid carrying a current produces a magnetic field B along its axis. If the current is doubled and the number of turns per cm is halved, the new value of magnetic field will be equal to [25 June, 2022 (Shift-II)]
- (a) B (b) $2B$
 (c) $4B$ (d) $\frac{B}{2}$
13. A charge particle is moving in a uniform magnetic field $(2\hat{i} + 3\hat{j})\text{T}$. If it has an acceleration of $(\alpha\hat{i} - 4\hat{j})\text{m/s}^2$, then the value of α is [26 July, 2022 (Shift-I)]
- (a) 3 (b) 6
 (c) 12 (d) 2
14. A wire A, bent in the shape of an arc of a circle, carrying a current of 2 A and having radius 2 cm and another wire B, also bent in the shape of arc of a circle, carrying a current of 3 A and having radius of 4 cm, are placed as shown in the figure. The ratio of the magnetic fields due to the wires A and B at the common centre O is

[04 Sep, 2020 (Shift-I)]

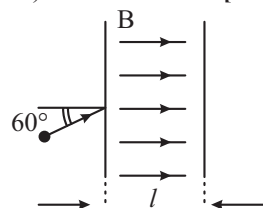


- (a) 2 : 5
 (b) 6 : 5
 (c) 4 : 6
 (d) 6 : 4

15. Two ions having same mass have charges in the ratio 1 : 2. They are projected normally in a uniform magnetic field with their speeds in the ratio 2 : 3. The ratio of the radii of their circular trajectories is:

[25 July, 2021 (Shift-II)]

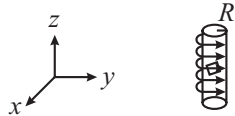
- (a) 3 : 1 (b) 2 : 3
 (c) 4 : 3 (d) 1 : 4
16. A proton, a deuteron and an α -particle are moving with same momentum in a uniform magnetic field. The ratio of magnetic forces acting on them is _____ and their speed is _____, in the ratio. [25 Feb, 2021 (Shift-I)]
- (a) 4 : 2 : 1 and 2 : 1 : 1 (b) 2 : 1 : 1 and 4 : 2 : 1
 (c) 1 : 2 : 4 and 1 : 1 : 2 (d) 1 : 2 : 4 and 2 : 1 : 1
17. A particle of mass m and charge q has an initial velocity $\vec{v} = v_0\hat{j}$. If an electric field $\vec{E} = E_0\hat{i}$ and magnetic field $\vec{B} = B_0\hat{i}$ act on the particle, its speed will double after a time [7 Jan, 2020 (Shift-II)]
- (a) $\frac{2mv_0}{qE_0}$
 (b) $\frac{3mv_0}{qE_0}$
 (c) $\frac{\sqrt{2}mv_0}{qE_0}$
 (d) $\frac{\sqrt{3}mv_0}{qE_0}$
18. The figure shows a region of length ' ℓ ' with a uniform magnetic field of 0.3 T in it and a proton entering the region with velocity $4 \times 10^5 \text{ ms}^{-1}$ making an angle 60° with the field. If the proton completes 10 revolution by the time it cross the region shown, ' ℓ ' is close to (mass of proton = $1.67 \times 10^{-27} \text{ kg}$, charge of the proton = $1.6 \times 10^{-19} \text{ C}$) [2 Sep, 2020 (Shift-II)]



- (a) 0.44 m (b) 0.11 m
 (c) 0.88 m (d) 0.22 m
19. A beam of protons with speed $4 \times 10^5 \text{ m/s}$ enters a uniform magnetic field of 0.3 T at an angle of 60° to the magnetic field. The pitch of the resulting helical path of protons is close to (Mass of the proton = $1.67 \times 10^{-27} \text{ kg}$, charge of the proton = $1.69 \times 10^{-19} \text{ C}$) [2 Sep, 2020 (Shift-I)]
- (a) 2 cm (b) 12 cm
 (c) 5 cm (d) 4 cm

20. An electron gun is placed inside a long solenoid of radius R on its axis. The solenoid has n turns/length and carries a current I . The electron gun shoots an electron along the radius of the solenoid with speed v . If the electron does not hit the surface of the solenoid, maximum possible value of v is (all symbols have their standard meaning)

[9 Jan, 2020 (Shift-II)]



- (a) $\frac{e\mu_0 nIR}{2m}$ (b) $\frac{e\mu_0 nIR}{m}$
 (c) $\frac{2e\mu_0 nIR}{m}$ (d) $\frac{e\mu_0 nIR}{4m}$
21. In an experiment, electrons are accelerated, from rest, by applying, a voltage of 500 V. Calculate the radius of the path if a magnetic field 100 mT is then applied. [Charge of the electron = 1.6×10^{-19} C Mass of the electron = 9.1×10^{-31} kg]
- [11 Jan, 2019 (Shift-I)]
- (a) 7.5×10^{-3} m (b) 7.5×10^{-2} m
 (c) 7.5 m (d) 7.5×10^{-4} m
22. A proton, an electron, and a Helium nucleus, have the same energy. They are in circular orbitals in a plane due to magnetic field perpendicular to the plane. Let r_p , r_e and r_{He} be their respective radii, then,

[10 April, 2019 (Shift-I)]

- (a) $r_e > r_p > r_{He}$ (b) $r_e < r_p < r_{He}$
 (c) $r_e < r_p = r_{He}$ (d) $r_e > r_p = r_{He}$
23. The region between $y = 0$ and $y = d$ contains a magnetic field $\vec{B} = B\hat{z}$. A particle of mass m and charge q enters the region with a velocity $\vec{v} = v\hat{i}$. If $d = \frac{mv}{2qB}$, the acceleration of the charged particle at the point of its emergence at the other side is:

[11 Jan, 2019 (Shift-II)]

- (a) $\frac{qvB}{m} \left(\frac{1}{2}\hat{i} - \frac{\sqrt{3}}{2}\hat{j} \right)$ (b) $\frac{-qvB}{m} \left(\frac{\sqrt{3}}{2}\hat{i} + \frac{1}{2}\hat{j} \right)$
 (c) $\frac{qvB}{m} \left(\frac{-\hat{j} + \hat{i}}{\sqrt{2}} \right)$ (d) $\frac{qvB}{m} \left(\frac{\hat{j} + \hat{i}}{\sqrt{2}} \right)$
24. Two long straight wires P and Q carrying equal current of $10A$ each were kept parallel to each other at a distance of 5cm. Magnitude of magnetic force experienced by 10cm length of wire P is F_1 . If distance between the wires is halved and currents on them are doubled, force F_2 on 10 cm length of wire P will be:

[24 Jan, 2023 (Shift-I)]

- (a) $8F_1$ (b) $10F_1$
 (c) $F_1/8$ (d) $F_1/10$

25. A particle of mass m and charge q is in an electric and magnetic field given by:

$$\vec{E} = 2\hat{i} + 3\hat{j}; \vec{B} = 4\hat{j} + 6\hat{k}$$

The charged particle is shifted from the origin to the point $P(x = 1; y = 1)$ along a straight path. The magnitude of the total work done is:

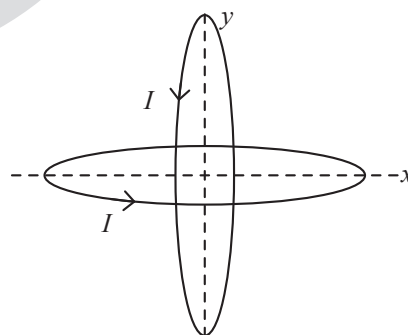
[11 Jan, 2019 (Shift-II)]

- (a) $(0.35)q$
 (b) $5q$
 (c) $(2.5)q$
 (d) $(0.15)q$
26. The current sensitivity of moving coil galvanometer is increased by 25%. This increase is achieved only by changing in the number of turns of coils and area of cross section of the wire while keeping the resistance of galvanometer coil constant. The percentage change in the voltage sensitivity will be:
- [11 April, 2023 (Shift-I)]
- (a) +25% (b) -50%
 (c) Zero (d) -25%

Integer Type Questions

27. Two identical circular wires of radius 20 cm and carrying current $\sqrt{2}A$ are placed in perpendicular planes as shown in figure. The net magnetic field at the centre of the circular wire is $\times 10^{-8} T$. (Take $\pi = 3.14$

[6 April, 2023 (Shift-I)]



28. A charge particle of $2 \mu C$ accelerated by a potential difference of 100 V enters a region of uniform magnetic field of magnitude 4 mT at right angle to the direction of field. The charge particle completes semicircle of radius 3 cm inside magnetic field. The mass of the charge particle is $\times 10^{-18} \text{ kg}$.

[1 Feb, 2023 (Shift-I)]

29. A proton with a kinetic energy of 2.0 eV moves into a region of uniform magnetic field of magnitude $\frac{\pi}{2} \times 10^{-3} T$. The angle between the direction of magnetic field and velocity of proton is 60° . The pitch of the helical path taken by the proton is _____ cm.
(Take, mass of proton = 1.6×10^{-27} kg and Charge on proton = 1.6×10^{-19} C). [6 April, 2023 (Shift-II)]

30. A straight wire AB of mass 40 g and length 50 cm is suspended by a pair of flexible leads in uniform magnetic field of magnitude 0.40 T as shown in the figure. The magnitude of the current required in the wire to remove the tension in the supporting leads is _____ A. (Take $g = 10 \text{ ms}^{-2}$). [13 April, 2023 (Shift-II)]



ANSWER KEY

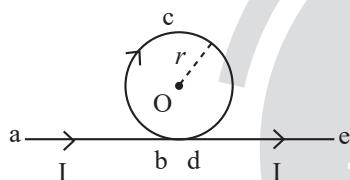
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|---------|---------|---------|---------|---------|---------|-----------|-----------|----------|---------|
| 1. (d) | 2. (c) | 3. (c) | 4. (c) | 5. (b) | 6. (d) | 7. (a) | 8. (c) | 9. (d) | 10. (a) |
| 11. (d) | 12. (a) | 13. (b) | 14. (b) | 15. (c) | 16. (a) | 17. (d) | 18. (a) | 19. (d) | 20. (a) |
| 21. (d) | 22. (c) | 23. (b) | 24. (a) | 25. (b) | 26. (a) | 27. [628] | 28. [144] | 29. [40] | 30. [2] |

EXPLANATIONS

1. (d) Magnetic field at point O due to current in BC and ET is

$$B_0 = \frac{\mu_0 I}{4\pi \frac{a}{2}} + \frac{\mu_0 I}{4\pi \frac{a}{2}} = \frac{\mu_0 I}{\pi a}$$

2. (c) For option (A)



$$B_{ab} = \frac{\mu_0 I}{4\pi r} \text{ (outwards)}$$

$$B_{bcd} = \frac{\mu_0 I}{4\pi r} (2\pi) \text{ (inwards)}$$

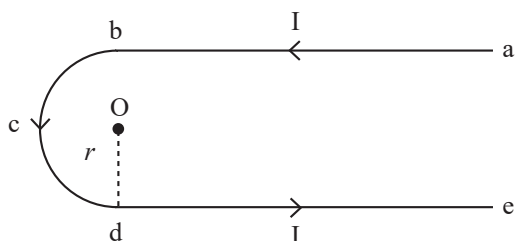
$$B_{de} = \frac{\mu_0 I}{4\pi r} \text{ (outwards)}$$

Hence magnetic field at O is

$$B_0 = -\frac{\mu_0 I}{4\pi r} + \frac{\mu_0 I}{4\pi r} (2\pi) - \frac{\mu_0 I}{4\pi r}$$

$$B_0 = \frac{\mu_0 I}{2\pi r} (\pi - 1) \quad \dots \text{option (III)}$$

For Option (B)



$$B_{ab} = \frac{\mu_0 I}{4\pi r} \text{ (outwards)}$$

$$B_{bcd} = \frac{\mu_0 I}{4\pi r} (\pi) \text{ (outwards)}$$

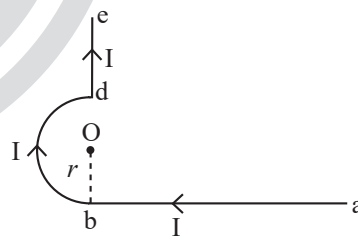
$$B_{de} = \frac{\mu_0 I}{4\pi r} \text{ (outwards)}$$

Hence magnetic field at O is

$$B_0 = \frac{\mu_0 I}{4\pi r} + \frac{\mu_0 I}{4\pi r} (\pi) + \frac{\mu_0 I}{4\pi r}$$

$$B_0 = \frac{\mu_0 I}{4\pi r} (\pi + 2) \dots \text{option (I)}$$

For option (C)



$$B_{ab} = \frac{\mu_0 I}{4\pi r} \text{ (inwards)}$$

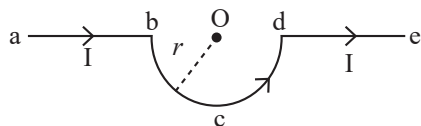
$$B_{bcd} = \frac{\mu_0 I}{4\pi r} (\pi) \text{ (inwards)}$$

$$B_{de} = 0 \text{ (at point O)}$$

Hence magnetic field at O is

$$B_0 = \frac{\mu_0 I}{4\pi r} (1 + \pi) \dots \text{option (IV)}$$

For option (D)



$$B_{ab} = 0 \text{ (at point } O\text{)}$$

$$B_{bcd} = \frac{\mu_0 I}{4\pi r} (\pi) \text{ (outwards)}$$

$$B_{de} = 0 \text{ (outwards)}$$

$$\text{Hence magnetic field at } O \text{ is } B_0 = \frac{\mu_0 I}{4r} \dots \text{option (II)}$$

$$3. (c) B_C = \frac{\mu_0 I}{2r}, B_a = \frac{\mu_0 I r^2}{2(x^2 + r^2)^{3/2}}$$

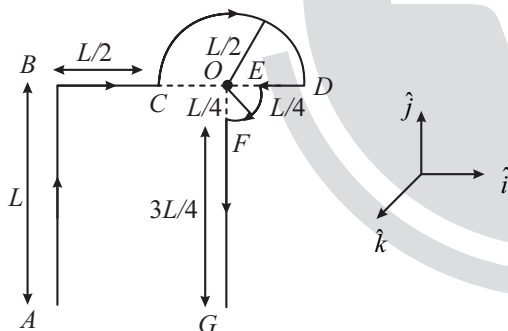
$$\text{At } x = \frac{r}{2}$$

$$B_a = \frac{\mu_0 I r^2}{2\left(\frac{r^2}{4} + r^2\right)^{3/2}}$$

$$= \frac{\mu_0 I r^2}{2\left(\frac{5}{4}r^2\right)^{3/2}} = \frac{\mu_0 I}{2r} \left(\frac{4}{5}\right)^{3/2}$$

$$= \frac{\mu_0 I}{2r} \left(\frac{2}{\sqrt{5}}\right)^3 = \left(\frac{2}{\sqrt{5}}\right)^3 B$$

4. (c)



$$\vec{B}_{\text{Net}} = \vec{B}_{AB} + \vec{B}_{BC} + \vec{B}_{CD} + \vec{B}_{DE} + \vec{B}_{EF} + \vec{B}_{FG}$$

$$\vec{B}_{BC} = \vec{B}_{DE} = \vec{B}_{FG} = 0$$

$$\vec{B}_{AB} = \frac{\mu_0 I}{4\pi L} \sin 45^\circ [-\hat{k}]$$

$$\vec{B}_{CD} = \frac{\mu_0 I}{4\left(\frac{L}{2}\right)^2} [-\hat{k}]$$

$$\vec{B}_{EF} = \frac{\mu_0 I}{8\left(\frac{L}{4}\right)^2} [-\hat{k}]$$

5. (b) Number of turns = N

Inner radius = a , outer radius = b ,

Consider an elementary portion of radius r and thickness dr .

Number of turns in elementary portion,

$$dN = \frac{N}{2\pi(b-a)} \cdot 2\pi dr = \frac{N}{(b-a)} dr$$

$$\therefore (dB) \text{ at centre} = \frac{\mu_0 I dN}{2r} \Rightarrow B = \int_a^b \frac{\mu_0 I}{2r} \times \left(\frac{N}{b-a}\right)$$

$$\Rightarrow B = \frac{\mu_0 I N}{2(b-a)} \ln\left(\frac{b}{a}\right)$$

6. (d) Step 1. Given data

A hairpin is made by bending a long current-carrying wire.

Step 2. Finding the magnitude of the magnetic field at the point P :

$$\text{Magnetic field due to each of the straight wire} = \frac{\mu_0 I}{4\pi r}$$

$$\text{Magnetic field due to semicircular arc} = \frac{\mu_0 I}{4r}$$

Thus, the total magnetic field at a point P is given by

$$B = \frac{\mu_0 I}{4\pi r} + \frac{\mu_0 I}{4\pi r} + \frac{\mu_0 I}{4r}$$

$$B = \frac{\mu_0 I}{2\pi r} + \frac{\mu_0 I}{4r}$$

$$B = \frac{\mu_0 I}{4\pi r} (2 + \pi)$$

7. (a) $m_p = m$

$$q_p = q$$

$$K_p = qV = K$$

$$m_\alpha = 4m \quad q_\alpha = 2q \quad K_\alpha = 2qV = 2K$$

Radius of circular path,

$$r = \frac{mv}{qB} = \frac{\sqrt{2Km}}{qB} = \sqrt{\frac{2mV}{qB^2}} \Rightarrow \frac{r_p}{r_\alpha} = \frac{1}{\sqrt{2}}$$

$$8. (c) B = 2 \left[\frac{\mu_0 i}{4\pi d} (\cos \theta_1 - \cos \theta_2) \right]$$

$$B = 10^{-4}$$

$$\mu_0 = 4\pi \times 10^{-7}, \theta_1 = 90^\circ, \theta_2 = 180^\circ$$

$$d = 4 \times 10^{-2}$$

$$\Rightarrow = 20 \text{ A (into the page)}$$

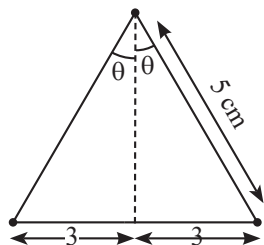
9. (d) $B = \frac{\mu_0 I}{4\pi d} 2\sin\theta$

$d = 4 \text{ cm}$

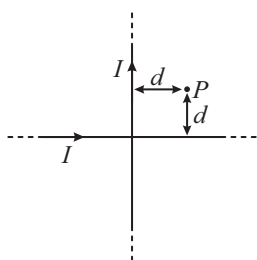
$\sin\theta = \frac{3}{5}$

$B = 10^{-7} \times \frac{5 \times 2}{4 \times 10^{-2}} \left(\frac{3}{5}\right)$

$B = 1.5 \times 10^{-5} \text{ T}$



10. (a)



Magnetic field at point P

$\vec{B}_{net} = \frac{\mu_0 I}{2\pi d} (-\hat{k}) + \frac{\mu_0 I}{2\pi d} (\hat{k}) = 0$

11. (d) Magnetic field at centre due to current carrying loop subtending angle θ at the centre, $B = \frac{\mu_0 I}{4\pi r} \theta$.

Hence, θ is in radian.

$B = \left(\frac{\mu_0}{4\pi}\right) \times 10 \left(\frac{1}{3 \times 10^{-2}} - \frac{1}{5 \times 10^{-2}}\right) \frac{\pi}{4}$

$\Rightarrow B = \frac{\pi}{40} \times 10^{-4} = \frac{\pi}{4} \times 10^{-5} \approx 10^{-5}$

12. (a) We know that, $B = n\mu_0 I$

According to equations $I' = 2I$

$n' = n/2$

$B' = \frac{n}{2} \mu_0 2I = B$

13. (b) $\vec{a} \cdot \vec{B} = 0$

$(\alpha\hat{i} - 4\hat{j}) \cdot (2\hat{i} + 3\hat{j}) = 0$

$\alpha(2) + (-4)(3) = 0$

$\alpha = \frac{12}{2} \Rightarrow \alpha = 6$

14. (b) $B_A = \frac{(\mu_0)(2) \frac{3(\pi/2)}{2\pi}}{(2)(2)}$... (i)

$B_B = \frac{(\mu_0)(3) \frac{5\pi/3}{2\pi}}{(2)(4)}$... (ii)

From (i) and (ii)

$\frac{B_A}{B_B} = \frac{6}{5}$

15. (c) Given, $\frac{q_1}{q_2} = \frac{1}{2}, \frac{v_1}{v_2} = \frac{2}{3}$

Radius of the circular trajectory in the magnetic field.

$r = \frac{mv}{qB}$

$\therefore \frac{r_1}{r_2} = \left(\frac{m_1}{m_2}\right) \left(\frac{q_2}{q_1}\right) \left(\frac{v_1}{v_2}\right)$

$\frac{r_1}{r_2} = 2 \times \frac{2}{3} = \frac{4}{3}$

16. (a)

Proton

α -particle

Deuteron

Speed, $v = P/m$

$v_1 = \frac{P}{m}, v_2 = \frac{P}{2m}, v_3 = \frac{P}{4m}$

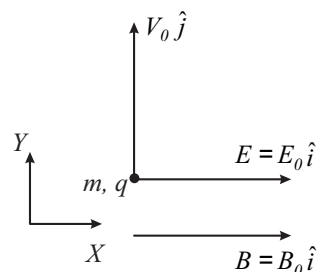
$\therefore v_1 : v_2 : v_3 = 4 : 2 : 1$

$F = qvB = \frac{qPB}{m}$

$F_1 = \frac{qPB}{m}, F_2 = \frac{q \times P \times B}{2m}, F_3 = \frac{2qPB}{4m}$

$\therefore F_1 : F_2 : F_3 = 4 : 2 : 2 = 2 : 1 : 1$

17. (d)



$$(2V_0)^2 = v_0^2 + v_x^2; v_x = \sqrt{3}v_0$$

$$\therefore \sqrt{3}v_0 = 0 + \frac{eE}{m}t \Rightarrow t = \frac{mv_0\sqrt{3}}{qE}$$

$$\begin{aligned} 18. (a) \ell &= 10 \times \text{pitch} \\ &= 10 \times v \cos 60^\circ \times \frac{2\pi m}{qB} \\ &= \frac{20 \times 3.14 \times 1.67 \times 10^{-27} \times 4 \times 10^5 \times \frac{1}{2}}{1.6 \times 10^{-19} \times 0.3} \\ &= 4.36 \times 10^{-1} \text{ m} \approx 0.44 \text{ m}. \end{aligned}$$

$$\begin{aligned} 19. (d) \text{Pitch} &= v \cos \theta \times \frac{2\pi m}{qB} \\ &= 4 \times 10^5 \times \frac{1}{2} \times \frac{2\pi \times 1.67 \times 10^{-27}}{1.69 \times 10^{-19} \times 0.3} = 4 \text{ cm} \end{aligned}$$

$$20. (a) R_{\max} = \frac{R}{2} = \frac{mv_{\max}}{e\mu_0 nI}$$

$$V_{\max} = \frac{Re\mu_0 nI}{2m}$$

$$\begin{aligned} 21. (d) \text{Radius, } r &= \frac{mv}{qB} = \frac{\sqrt{2mqv}}{qB} \\ r &= \frac{\sqrt{\frac{2mv}{q}}}{B} = \frac{\sqrt{\frac{2 \times 9.1 \times 10^{-31} \times 500}{1.6 \times 10^{-19}}}}{100 \times 10^{-3}} \\ r &= 7.5 \times 10^{-4} \text{ m} \end{aligned}$$

$$22. (c) r = \frac{mv}{qB} = \frac{\sqrt{2mK}}{qB}$$

$$\Rightarrow r_{He} = r_p > r_e$$

$$23. (b) \hat{a} = -\left(\frac{d\sqrt{3}\hat{i} + d\hat{j}}{\sqrt{(d\sqrt{3})^2 + d^2}} \right) = \frac{-\sqrt{3}\hat{i} - \hat{j}}{2}$$

$$\therefore \text{Acceleration, } \vec{a} = \frac{F}{m} \hat{a} = \frac{qVB}{m} \left(\frac{-\sqrt{3}\hat{i} - \hat{j}}{2} \right)$$

24. (a) Force per unit length between two parallel straight

$$\text{current carrying wires} = \frac{\mu_0 i_1 i_2}{2\pi d}$$

$$\begin{aligned} \Rightarrow \frac{F_1}{F_2} &= \frac{\frac{\mu_0 (10)^2}{2\pi (5\text{cm})}}{\frac{\mu_0 (20)^2}{2\pi \left(\frac{5\text{cm}}{2}\right)}} = \frac{1}{8} \end{aligned}$$

$$\Rightarrow F_2 = 8F_1$$

25. (b) Net force of charge,

$$\begin{aligned} \vec{F}_{\text{net}} &= q\vec{E} + q(\vec{v} \times \vec{B}) = q\vec{E} + q|\vec{v}||\vec{B}|\sin 90^\circ \\ &= (2q\hat{i} + 3q\hat{j} + 0) \end{aligned}$$

$$\begin{aligned} \text{Work done } W &= \vec{F}_{\text{net}} \cdot \vec{S} = (2q\hat{i} + 3q\hat{j}) \cdot (\hat{i} + \hat{j}) \\ &= 2q + 3q = 5q \end{aligned}$$

26. (a) Current sensitivity is the deflection (ϕ) per unit current

$$(i) \text{ i.e., } \frac{\phi}{I} = \frac{NAB}{K}$$

$$\text{Voltage sensitivity, } \frac{\phi}{V} = \frac{\phi}{IR_g} = \frac{\text{current sensitivity}}{R_g}$$

Voltage sensitivity $\propto$ current sensitivity (when R_g is constant)

So percentage change in V_s is 25%

$$27. [628] B_y = |B_x| = \frac{\mu_0 I}{2r}$$

$$= \frac{4\pi \times 10^{-7}}{2 \times 0.2} \times \sqrt{2} T$$

Net magnetic field at the centre,

$$B_{\text{net}} = B\sqrt{2} = \frac{4\pi \times 10^{-7} \times \sqrt{2}}{2 \times 0.2} \times \sqrt{2} T = 2\pi \times 10^{-6} N$$

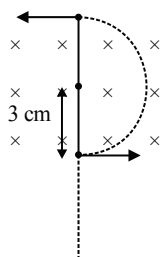
$$= 200\pi \times 10^{-8} T$$

$$= 2 \times 314 \times 10^{-8} T$$

$$= 628 \times 10^{-8} T$$

$$28. [144] r = \frac{mv}{qB} = \frac{\sqrt{2km}}{qB}, m = \frac{r^2 q^2 B^2}{2k}$$

$$m = \frac{r^2 q^2 B^2}{2k}$$



$$m = \frac{1}{100} \times \frac{3}{100} \times 2 \times 2 \times 4 \times 10^{-3} \times 4 \times 10^{-3} \times 10^{-12}$$

$$2 \times (100) \times 10^{-6}$$

$$\therefore m = 144 \times 10^{-18} \text{ kg}$$

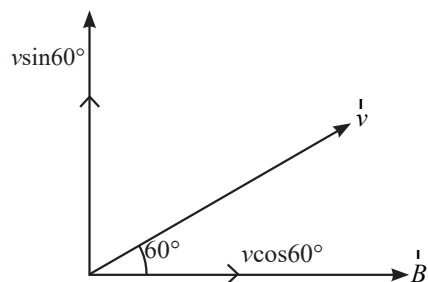
29. [40] $B = \frac{\pi}{2} \times 10^{-3}$

$$v = \sqrt{\frac{2KE}{m}} = \sqrt{\frac{2 \times 2 \times 1.6 \times 10^{-19}}{1.6 \times 10^{-27}}}$$

$$v = 2 \times 10^4 \text{ m/sec}$$

$$\text{Time period of one rotation} = \frac{2\pi m}{eB}$$

$$T = \frac{2\pi \times 1.6 \times 10^{-27}}{1.6 \times 10^{-19} \times \frac{\pi}{2} \times 10^{-3}} = 4 \times 10^{-5} \text{ sec}$$



$$\text{Pitch} = (v \cos 60^\circ)T$$

$$\text{Pitch} = 2 \times 10^4 \times \frac{1}{2} \times 4 \times 10^{-5} = 40 \text{ cm}$$

30. [2] For equilibrium, Gravitational force acting on the wire = Magnetic force on the wire

$$Mg = IlB$$

$$I = \frac{mg}{lB} = \frac{40 \times 10^{-3} \times 10}{50 \times 10^{-2} \times 0.4} = 2 \text{ A}$$